Exp 13-Pandas Program to Identify Missing Values and Display True or False

Aim

To write a Pandas program to detect missing values in a given DataFrame and display the result as True or False.

Procedure

1. Import the Pandas library.
2. Create a DataFrame containing the given data.
3. Use NaN to represent missing values.
4. Apply the `isnull()` function to the DataFrame.
5. Display the output, where:
    - True indicates a missing value.
    - False indicates a non-missing value.

CODE

```
import pandas as pd
import numpy as np

# Create the DataFrame

data = {
    'ord_no': [70001, np.nan, 70002, 70004, np.nan,
               70005, np.nan, 70010, 70003, 70012,
               np.nan, 70013],

    'purch_amt': [150.50, 270.65, 65.26, 110.50, 948.50,
                  2400.60, 5760.00, 1983.43, 2480.40,
                  250.45, 75.29, 3045.60],

    'ord_date': ['2012-10-05', '2012-09-10', np.nan,
                 '2012-08-17', '2012-09-10', '2012-07-27',
                 '2012-09-10', '2012-10-10', '2012-10-10',
                 '2012-06-27', '2012-08-17', '2012-04-25'],

    'customer_id': [3002, 3001, 3001, 3003, 3002,
                    3001, 3001, 3004, 3003, 3002,
                    3001, 3001],

    'salesman_id': [5002, 5003, 5001, np.nan, 5002,
                    5001, 5001, np.nan, 5003, 5002,
                    5003, np.nan]
}

df = pd.DataFrame(data)

print("Original DataFrame:")
print(df)

print("\nMissing values (True/False):")
print(df.isnull())
```

OUTPUT

```
Original DataFrame:
     ord_no  purch_amt    ord_date  customer_id  salesman_id
0   70001.0     150.50  2012-10-05         3002       5002.0
1       NaN     270.65  2012-09-10         3001       5003.0
2   70002.0      65.26         NaN         3001       5001.0
3   70004.0     110.50  2012-08-17         3003          NaN
4       NaN     948.50  2012-09-10         3002       5002.0
5   70005.0    2400.60  2012-07-27         3001       5001.0
6       NaN    5760.00  2012-09-10         3001       5001.0
7   70010.0    1983.43  2012-10-10         3004          NaN
8   70003.0    2480.40  2012-10-10         3003       5003.0
9   70012.0     250.45  2012-06-27         3002       5002.0
10      NaN      75.29  2012-08-17         3001       5003.0
11  70013.0    3045.60  2012-04-25         3001          NaN

Missing values (True/False):
    ord_no  purch_amt  ord_date  customer_id  salesman_id
0    False      False     False        False        False
1     True      False     False        False        False
2    False      False      True        False        False
3    False      False     False        False         True
4     True      False     False        False        False
5    False      False     False        False        False
6     True      False     False        False        False
7    False      False     False        False         True
8    False      False     False        False        False
9    False      False     False        False        False
10    True      False     False        False        False
11   False      False     False        False         True
```

Exp 14-Finding and Replacing Missing Values in a DataFrame Using Pandas

Aim

To write a Pandas program to find and replace the missing values in a given DataFrame that do not contain any valuable information

Procedure

1. Import the Pandas and NumPy libraries.
2. Create a DataFrame with the given data.
3. Replace special symbols such as ?, `--`, and NaN with a common missing value (NaN).
4. Use the `replace()` function to identify missing values.
5. Replace all missing values with `"Not Available"`.
6. Display the updated DataFrame.

CODE

```
import pandas as pd
import numpy as np

data = {
    'ord_no': [70001, np.nan, 70002, 70004, np.nan,
               70005, '--', 70010, 70003, 70012,
               np.nan, 70013],

    'purch_amt': [150.5, 270.65, 65.26, 110.5, 948.5,
                  2400.6, 5760, '?', 12.43, 2480.4,
                  250.45, 3045.6],

    'ord_date': ['?', '2012-09-10', np.nan, '2012-08-17',
                 '2012-09-10', '2012-07-27', '2012-09-10',
                 '2012-10-10', '2012-10-10', '2012-06-27',
                 '2012-08-17', '2012-04-25'],

    'customer_id': [3002, 3001, 3001, 3003, 3002,
                    3001, 3001, 3004, '--', 3002,
                    3001, 3001],

    'salesman_id': [5002, 5003, '?', 5001, np.nan,
                    5002, 5001, '?', 5003, 5002,
                    5003, '--']
}

df = pd.DataFrame(data)

print("Original DataFrame:")
print(df)

df.replace(['?', '--'], np.nan, inplace=True)

df.fillna("Not Available", inplace=True)

print("\nUpdated DataFrame:")
print(df)
```

OUTPUT

```
Original DataFrame:
   ord_no purch_amt    ord_date customer_id salesman_id
0   70001     150.5           ?        3002        5002
1     NaN    270.65  2012-09-10        3001        5003
2   70002     65.26         NaN        3001           ?
3   70004     110.5  2012-08-17        3003        5001
4     NaN     948.5  2012-09-10        3002         NaN
5   70005    2400.6  2012-07-27        3001        5002
6      --      5760  2012-09-10        3001        5001
7   70010         ?  2012-10-10        3004           ?
8   70003     12.43  2012-10-10          --        5003
9   70012    2480.4  2012-06-27        3002        5002
10    NaN    250.45  2012-08-17        3001        5003
11  70013    3045.6  2012-04-25        3001          --
```

```
Updated DataFrame:
           ord_no      purch_amt       ord_date    customer_id    salesman_id
0         70001.0          150.5  Not Available         3002.0         5002.0
1   Not Available         270.65     2012-09-10         3001.0         5003.0
2         70002.0          65.26  Not Available         3001.0  Not Available
3         70004.0          110.5     2012-08-17         3003.0         5001.0
4   Not Available          948.5     2012-09-10         3002.0  Not Available
5         70005.0         2400.6     2012-07-27         3001.0         5002.0
6   Not Available         5760.0     2012-09-10         3001.0         5001.0
7         70010.0  Not Available     2012-10-10         3004.0  Not Available
8         70003.0          12.43     2012-10-10  Not Available         5003.0
9         70012.0         2480.4     2012-06-27         3002.0         5002.0
10  Not Available         250.45     2012-08-17         3001.0         5003.0
11        70013.0         3045.6     2012-04-25         3001.0  Not Available
```